A Multi-Agent Large Language Model Framework for Explainable Stock Research and Investment Candidate Selection

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***Abstract*—Before an analyst can form a view on a stock, they usually have to piece together price movements, company fundamentals, recent news, and downside risks from several different places. Doing this by hand is slow, and the results can vary from one analyst or one day to the next. This paper describes the design of a multi-agent research assistant that splits this work among specialized large language model (LLM) agents. One agent proposes candidate companies, another gathers financial and news evidence, and a third weighs that evidence to reach a reasoned Buy, Hold, or Avoid-style assessment. An orchestration agent coordinates the stages, search and financial-data tools supply the evidence, and a final step turns the assessment into a portable PDF report that can be sent by email. The system produces advisory research only; it never places trades. We describe the architecture, workflow, data handling, and an evaluation plan. Because a controlled backtest and a quantitative user study have not yet been carried out, we make no claims about predictive accuracy or investment returns. The framework is a decision-support prototype and should not be read as financial advice.**

***Index Terms*—multi-agent systems, large language models, financial research, stock screening, explainable AI, decision support, CrewAI**

I. Introduction

A recent share price tells an analyst very little on its own. A defensible view of a company usually draws on price and volume history, revenue and earnings trends, profitability, valuation, leverage, cash flow, company announcements, market news, and a clear picture of what could go wrong. Collecting these signals from scattered sources takes time, and the effort multiplies when several companies must be compared.

Large language models (LLMs) are good at reading unstructured text and can call external tools to fetch evidence. Left to work alone, however, a single model can blur its responsibilities, miss information that contradicts its story, or state things it cannot support. The ReAct paradigm, which alternates language-model reasoning with actions that return observations from tools, gives a sound conceptual basis for tool-using agents [1]. Splitting the job across several agents with narrower roles builds on that idea and makes each intermediate result easier to inspect.

This paper presents the Stock Picker Agent, a multi-agent research workflow built around company discovery, financial research, candidate selection, report generation, and email delivery. Its purpose is to help a user organize evidence and compare candidate companies. It does not predict future prices, and its output is not a replacement for professional financial advice.

The main contributions are as follows:

- A modular workflow that keeps candidate discovery, financial research, and final stock assessment as separate stages.
- A report structure centered on evidence, which records supporting signals, risks, and data that could not be obtained.
- An end-to-end design that carries research output through to PDF generation and email delivery.
- An evaluation protocol covering factual grounding, completeness, reproducibility, latency, and recommendation quality, defined without inventing any performance results.

II. Related Work

A. Tool-Using Language Models

Yao et al. proposed ReAct, in which a model interleaves reasoning steps with external actions so that it can revise its understanding of a task from what the tools return [1]. This pattern fits financial research well, since company information lives in news pages, financial statements, and market-data services. Access to tools does not guarantee correct answers, though. Retrieved material may be outdated, partial, or self-contradictory, so the outputs still need to be checked.

B. Multi-Agent Task Decomposition

Multi-agent LLM workflows hand distinct responsibilities to specialized agents and then combine what they produce. Here, discovery, financial analysis, and final selection are kept apart so that no single prompt carries the whole burden and each stage can be examined on its own. The system uses CrewAI for orchestration [3]. The framework's role is purely operational coordination; it is not evidence that the resulting recommendations make money.

C. Financial Decision Support

Classical portfolio theory treats risk and return together instead of looking at expected return in isolation [2]. The

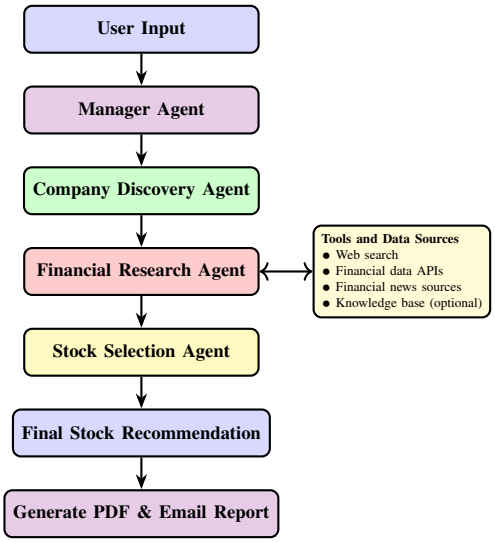


Fig. 1. Workflow of the proposed multi-agent Stock Picker Agent.

prototype described here is much narrower than a portfolio optimizer. It produces a research assessment for individual candidate stocks and makes no claim about optimal portfolio allocation. A dedicated risk-analysis stage is included so that favorable narratives are always read next to the adverse signals.

D. Research Gap and Motivation

Many lightweight stock-research prototypes rely on one LLM response or a digest of recent headlines. The workflow proposed here instead separates discovery, financial investigation, risk review, and final synthesis, and then packages the outcome as a report. This is a contribution to architecture. Whether it is useful in practice has to be shown through systematic evaluation, not assumed because several agents are involved.

III. System Overview

The system takes a research request, which may be a market or sector focus, or a single ticker when the workflow is configured for one company. A manager (orchestrator) agent controls the order of tasks and passes intermediate results to the agents that need them. Fig. 1 shows the overall workflow.

A. Manager / Orchestrator

The manager receives the research objective, hands out subtasks, and gathers the intermediate results. It also has to keep the identity of each company consistent from stage to stage, and to make sure the final report rests on what the research agents actually produced instead of an unrelated free-form answer.

B. Trending Company Discovery Agent

This agent proposes candidate companies from web-search results and other market-related information. For each candidate it should return the company name, ticker, the reason it was included, source links, and the retrieval date where available. A company that is widely discussed or trending counts only as a discovery signal. It is not evidence that the stock is undervalued or a suitable investment.

C. Financial Research Agent

The financial researcher gathers whatever market and fundamental data is available, such as recent price performance, revenue and earnings trends, profitability, valuation indicators, debt, and cash flow. A web-search tool like Serper [5] and financial-data sources such as Yahoo Finance, accessed for example through the `yfinance` library [4], can be connected when configured. Availability differs by company, exchange, data source, and API access. When a value cannot be found, it must be labeled as unavailable and not estimated.

D. Stock Selection and Risk Review

The selection stage compares the collected evidence and builds a structured assessment that covers positive signals, negative signals, and uncertainty. The output may carry labels such as Buy, Hold, or Avoid. These are heuristic research labels. They must not be presented as calibrated probabilities or guaranteed outcomes unless they have been validated on a defined dataset. The agent should explain the reasoning behind its assessment and point out any important data that is missing or conflicting.

E. Report Generation and Delivery

Once the synthesis is complete, the system formats the result as a PDF report and can send it through an email step. The report should name the company and ticker, timestamp the research, summarize the investment thesis, list the supporting evidence and the risks, cite its sources, and carry a disclaimer. Email delivery is a convenience for presentation and workflow. It does nothing to validate the recommendation itself.

IV. Methodology

A. Workflow

The proposed process runs through seven stages:

1) **Input normalization:** parse the request (market, sector, or ticker) and fix the scope of the analysis.
2) **Candidate discovery:** retrieve candidate companies and record why each one entered the research set.
3) **Evidence collection:** pull market, fundamental, and news information through the configured external tools.
4) **Financial analysis:** summarize the available financial indicators and the direction they have moved over the relevant window.
5) **Risk review:** identify adverse news, business risks, valuation concerns, leverage, and gaps in the evidence.
6) **Final synthesis:** compare the signals, produce a carefully qualified assessment, and keep the supporting evidence attached.
7) **Report delivery:** generate the PDF and, if enabled, email it to the intended recipient.

B. Evidence Representation

To keep the analysis traceable, every material claim in the final report should be stored as a tuple

$$e_i = (c_i,\, s_i,\, t_i,\, d_i,\, q_i), \tag{1}$$

where c_i is the claim or observation, s_i its source, t_i the time at which it was retrieved, d_i the date the underlying data refers to, and q_i a note on quality or availability. This representation is a design recommendation and should be built into the system's data structures if it is not already there.

The report should keep three things clearly apart: (i) facts taken directly from sources, (ii) the agent's interpretation of those facts, and (iii) information that is missing. If sources disagree, the disagreement should be shown to the reader and not resolved quietly. Claims drawn from news should come with a source URL and date whenever possible.

C. Candidate Assessment

A transparent scoring rubric can help organize the evidence, but its weights have to be chosen and validated beforehand, not tuned after looking at outcomes. As a starting design, the system reports four qualitative dimensions separately:

- **Fundamental condition:** the direction and consistency of reported financial indicators.
- **Market behavior:** recent price and volume context over a stated period.
- **News and catalysts:** relevant positive and negative developments, with source dates.
- **Risk and uncertainty:** valuation concerns, debt, business exposure, conflicting reports, and missing data.

These dimensions should stay individually visible in the report. If a single aggregate score is added later, its formula, scale, weights, and validation procedure must be documented. This paper claims no numerical scoring formula and no learned predictive model, since verified implementation details and evaluation data were not available.

D. Reliability Controls

We recommend the following controls for implementation and evaluation:

- Check the ticker and company identity before merging data from different sources.
- Attach retrieval timestamps, and never describe old data as current.
- Require a source for every material factual claim, and flag unsupported claims as unverified.
- Keep unavailable values marked as missing instead of filling them with model-generated guesses.
- Enforce a structured output schema and verify required report fields before PDF generation.
- Keep the final decision advisory, with human review required before any financial action.

TABLE I
PROPOSED EVALUATION PROTOCOL

Dimension	Measurement approach
Factual grounding	Manually audit sampled claims against the cited sources; report the rates of supported, contradicted, and unverifiable claims.
Completeness	Check that the required market, fundamental, news, risk, timestamp, and disclaimer fields are all present.
Source traceability	Measure the share of material claims that have a working source link and a retrieval date.
Reproducibility	Repeat identical requests and compare the retrieved evidence, intermediate outputs, and final labels.
Operational performance	Record end-to-end runtime, API failures, retry counts, and cost per completed report.
Recommendation quality	Use a time-separated historical evaluation with a predeclared universe, a benchmark, transaction costs, and no look-ahead leakage.
Usability	Have reviewers rate clarity, usefulness, and how easily the rationale can be traced to evidence.

V. IMPLEMENTATION CONSIDERATIONS

The project context points to CrewAI for multi-agent orchestration, Groq-hosted LLMs, Serper for web search, financial research tools, PDF generation, and email delivery. Exact model identifiers, prompt configurations, API versions, and data-provider settings should be taken from the actual repository and documented before final submission. They are deliberately left unspecified here because they can change between runs.

In practice, each stage can be written as an agent/task pair with explicit inputs and outputs. Intermediate results belong in a shared state object or a structured task result. The synthesis stage should build on the discovery and research outputs instead of repeating the whole research process without reference to earlier results. Failures caused by unavailable APIs, rate limits, or malformed responses should appear in the report or the logs, not disappear silently.

VI. EVALUATION PLAN

A research paper on this topic should assess both the quality of the generated analysis and the reliability of the workflow behind it. No benchmark results, completed backtest, or measured user-study results were available for this draft, so Table I lays out an evaluation plan and does not report experimental findings.

A. Recommended Historical Evaluation

If the system is ever used to make claims about investment performance, a backtest has to be run first. The stock universe and selection dates should be fixed in advance, and only information available at each decision date may be used. The experiment should also specify a holding period and rebalancing rule, and include transaction costs and a benchmark. Suitable

metrics include cumulative return, annualized return, volatility, maximum drawdown, and risk-adjusted performance, all compared with a simple baseline such as an equal-weight selection or a predefined market index. Survivorship bias and look-ahead bias need explicit attention.

VII. Discussion

The main strength of the design is task decomposition. Discovery and financial research pursue different goals, and a dedicated risk-review stage gives the system a chance to challenge an overly optimistic story. Packaging the rationale and its evidence into a single report also makes the output easier to use.

The architecture has limitations as well. Agent outputs depend on the quality, freshness, and coverage of external data. An LLM may misread financial terminology, summarize a source incorrectly, or give a plausible explanation that nothing supports. Several agents can also repeat the same underlying mistake, so the number of agents should never be treated as a guarantee of reliability. External APIs can change, impose rate limits, or leave out relevant companies and metrics. And a well-written report is not proof of predictive skill.

For these reasons, the system is best seen as a research assistant that helps structure and summarize information. It should not execute trades on its own, suggest certainty, or claim superior returns without controlled evidence.

VIII. Limitations and Future Work

This draft describes a project concept and a proposed technical methodology. It does not report a completed quantitative experiment, and the evidence on hand does not establish accuracy, profitability, latency, cost, or user satisfaction. Those quantities must be measured on the real implementation.

Planned future work includes: (1) a versioned, reproducible data-collection layer; (2) source-linked structured outputs enforced by schema; (3) a documented scoring rubric; (4) evaluation on historical data with strict temporal separation; (5) a comparison of the multi-agent workflow against a single-agent baseline; and (6) human review of factual support and report usefulness. Any move toward portfolio allocation would need separate modeling of diversification, risk tolerance, and constraints.

IX. Conclusion

This paper presented a multi-agent LLM framework that automates parts of stock research and candidate selection. The system coordinates company discovery, financial analysis, risk review, final synthesis, PDF report generation, and email delivery. Its central design principle is to give each stage a separate responsibility while preserving evidence, timestamps, uncertainty, and risks in the final output. The framework is a decision-support prototype and not a validated prediction engine. Empirical evaluation is still needed to show whether this architecture improves factual reliability, research efficiency, or investment outcomes.

Acknowledgment

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